

Exploring the Documentation of Delirium in Patients After Cardiac Surgery: A Retrospective Patient Record Study.

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Delirium is a common disorder for patients after cardiac surgery. Its manifestation and care can be examined through EHRs. The aim of this retrospective, comparative, and descriptive patient record study was to describe the documentation of delirium symptoms in the EHRs of patients who have undergone cardiac surgery and to explore how the documentation evolved between two periods (2005-2009 and 2015-2020). Randomly selected care episodes were annotated with a template, including delirium symptoms, treatment methods, and adverse events. The patients were then manually classified into two groups: nondelirious ($n = 257$) and possibly delirious ($n = 172$). The data were analyzed quantitatively and descriptively. According to the data, the documentation of symptoms such as disorientation, memory problems, motoric behavior, and disorganized thinking improved between periods. Yet, the key symptoms of delirium, inattention, and awareness were seldom documented. The professionals did not systematically document the possibility of delirium. Particularly, the way nurses recorded structural information did not facilitate an overall understanding of a patient's condition with respect to delirium. Information about delirium or proposed care was seldom documented in the discharge summaries. Advanced machine learning techniques can augment instruments that facilitate early detection, care planning, and transferring information to follow-up care.

Post-cardiac Surgery Delirium: When the Details Matter!

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Does Comprehensive Geriatric Assessment Reduce the Incidence of Postoperative Delirium? A Quasi-experimental Study in Older Adults Undergoing Transcatheter Aortic Valve Implantation.

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Postoperative delirium (POD) after transcatheter aortic valve implantation (TAVI) is frequent in older adults and associated with multiple negative outcomes including a higher mortality. We aimed to investigate whether a comprehensive geriatric assessment (CGA) prior to TAVI reduces the odds of POD and results in a positive change in self-care ability, intended to lay a foundation for future geriatric comanagement. We used a retrospective, single-center study with a quasi-experimental design enrolling patients aged 70 years and older undergoing CGA before elective TAVI, and a nonrandomized comparison group without preoperative CGA. Data on POD occurrence during the first 5 days after TAVI (primary outcome) and change in self-care ability index (SPI) between admission and discharge (secondary outcome) were collected from electronic health records and CGA data (exposure) by clinical assessment. To explore associations between (1) CGA and POD, and (2) CGA and SPI, multivariate logistic regression and linear regression models were applied adjusting for age, sex, BMI, and number of medications. Among 435 patients (mean age 81.0 ± 5.6 years, 43.6% women, median [IQR] SPI at baseline 40 [39, 40]

points), POD incidence was 14.3% in the CGA group vs 18.8% in the non-CGA group (P 0.219). Undergoing CGA before TAVI was not associated with the odds for POD (OR: 1.15; 95%CI: 0.65-2.04) or improved SPI (P 0.073). We observed no association of CGA prior to TAVI with POD incidence or postoperative self-care, highlighting the need for additional studies investigating the effect of POD preventive measures in older TAVI patients integrated into a comprehensive geriatric comanagement program.

Factors associated with delirium after cardiac surgery: A prospective cohort study.

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Delirium is a frequent complication after cardiac surgery and is associated with a higher incidence of morbidity and mortality and a prolonged hospital stay. However, knowledge of the variables involved in its occurrence is still limited; therefore, in this study, we evaluated the perioperative risk factors independently associated with this complication. This study was conducted in a referral tertiary care university hospital with a cardiovascular focus. A total of 311 consecutive adult patients undergoing any type of cardiac surgery were evaluated. The subjects were examined at regular intervals in the postoperative period using the Confusion Assessment Method for the Intensive Care Unit (CAM-ICU) tool. The incidence of postoperative delirium (PD) was 10%. Among the 18 pre-, intra- and postoperative variables evaluated, the logistic regression analysis showed that low education level, history of diabetes or stroke, type of surgery, prolonged extracorporeal circulation, or red blood cell transfusion in the intra- or postoperative period were independently associated with delirium after cardiac surgery. An increased body mass index was identified as a protective factor. The aforementioned risk factors are significantly and independently associated with the presentation of PD. Because some of these factors can be treated or avoided, the results of this study are highly relevant to reduce the risk of this complication and improve the care of patients undergoing cardiac surgery.

[Delirium after open cardiac surgery:systematic review of prevalence, risk factors and consequences].

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Delirium is a sudden and usually transient disturbance in consciousness, attention, cognition, perception and emotions. The pathophysiology is unknown but possible causes include neurotransmitter disturbances and inflammation reaction. Delirium is common in patients after open cardiac surgeries and can lead to serious consequences. Research shows that delirium is an underdiagnosed and undertreated problem. The purpose of this systematic review is to illuminate the prevalence, risk factors and outcome of postoperative delirium following open cardiac surgery. A systematic literature review from 2005-2013 was performed aiming to determine the prevalence, predisposing and precipitating factors and outcome after postoperative delirium following cardiac surgery. Web of Science, PubMed and Cinahl were searched. Findings of the systematic review shows that about one third of patients become delirious after cardiac surgery. Primary predisposing factors are advanced age, cognitive impairment, atrial fibrillation, depression and prior history of stroke. Among precipitating factors are pulmonary bypass, duration of mechanical ventilation, low cardiac output, respiratory failure, pneumonia, infections, blood cell transfusion and post-operative arrhythmias. Delirium causes prolonged hospital stay, reduced activity

and higher mortality. Delirium prevention includes reducing risk factors. Delirium is a common and serious complication of open cardiac surgery. Knowledge of risk factors of delirium and regular screening for symptoms of delirium are important to reduce prevalence and to facilitate diagnosis and treatment.

Early delirium after cardiac surgery: an analysis of incidence and risk factors in elderly (≥ 65 years) and very elderly (≥ 80 years) patients.

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Postoperative delirium is a common complication of cardiac surgery associated with increased mortality, morbidity, and long-term cognitive dysfunction. The aim of this study was to identify incidence and risk factors of delirium in elderly (≥ 65 years) and very elderly (≥ 80 years) patients undergoing major cardiac surgery. We performed a retrospective cohort analysis of prospectively collected data from a register of the cardiac surgery department of a tertiary referral university hospital between 2014 and 2016. Analysis was performed in two groups, ≥ 65 years and ≥ 80 years. We analyzed 1,797 patients ≥ 65 years, including 230 (7.24%) patients ≥ 80 years. Delirium was diagnosed in 21.4% (384/1,797) of patients above 65 years, and in 33.5% (77/230) of octogenarians. Early mortality did not differ between patients with and without delirium. Intensive care unit (ICU) stay ($p < 0.001$), hospital stay ($p < 0.001$), and intubation time ($p = 0.002$) were significantly longer in patients undergoing cardiac surgery ≥ 65 years with delirium. According to multivariable analysis, < 65 years, age (odds ratio [OR] 1.036, $p = 0.002$), low ejection fraction (OR 1.634, $p = 0.035$), diabetes (1.346, $p = 0.019$), and extracardiac arteriopathy (OR 1.564, $p = 0.007$) were found to be independent predictors of post-cardiac surgery delirium. Postoperative risk factors for developing delirium ≥ 65 years were atrial fibrillation (1.563, $p = 0.001$), postoperative pneumonia (OR 1.896, $p = 0.022$), elevated postoperative creatinine (OR 1.384, $p = 0.004$), and prolonged hospitalization (OR 1.019, $p = 0.009$). Patients above 65 years of age with postoperative delirium have poorer outcome and are more likely to have prolonged hospitalization and ICU stay, and longer intubation times, but 30-day mortality is not increased. In our study, eight independent risk factors for development of post-cardiac surgery delirium were age, low ejection fraction, diabetes, extracardiac arteriopathy, postoperative atrial fibrillation, pneumonia, elevated creatinine, and prolonged hospitalization time.

Delirium post-cardiac surgery: Incidence and associated factors.

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Post-operative delirium among cardiac surgery patients is a prevalent complication that associated with multiple negative outcomes. This study aimed to assess delirium incidence, associated factors, and outcomes for adult patients who underwent elective cardiac surgery. An exploratory prospective cohort design was used for this study. Delirium was diagnosed by the Confusion Assessment Method for the Intensive Care Unit. Incidence, preoperative, intraoperative, and post-operative variables for 245 patients during 3-month period were collected and analysed. Delirium developed in 9% ($n = 22$) of the sample. Patients with delirium were significantly older (mean age = 65.7, SD = 8.1), $t(243) = -3.66$, $P < .05$); had longer surgery time (mean time = 286.3, SD = 82.2), $t(243) = -2.25$, $P < .05$); received more blood post-surgery ($t(243) = -3.86$, $P < .05$); spent more time on mechanical ventilation ($t[21.6] = -2.2$, $P < .05$); had longer critical care unit stay ($t[21.7] = -4.0$, $P < 0.05$); and had longer hospitalization than patients without delirium. The risk factors associated with development of delirium were advanced age

and increased duration of surgery. Negative outcomes associated with delirium were increased duration of mechanical ventilation, increased volume of post-operative infused colloids and blood/products, increased critical care unit stay, and increased hospitalization. A multifactorial model for delirium risk factors should be considered to detect and work on potentially preventable delirium factors. Post-cardiac surgery delirium leads to longer mechanical ventilation time, increased ICU stay, and prolonged hospitalization. Delirium post-cardiac surgery is potentially preventable with appropriate identification of risk factors by nurses.

Predictors of delirium after cardiac surgery delirium: effect of beating-heart (off-pump) surgery.

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Despite improved outcomes after cardiac operations, postoperative delirium remains a common complication that is associated with increased morbidity and prolonged hospital stay. Univariate and multivariate predictors of postoperative delirium were determined from prospectively gathered data on 16,184 patients undergoing cardiac operations with cardiopulmonary bypass (conventional, $n = 14,342$) and without cardiopulmonary bypass (beating-heart surgery, $n = 1847$) between April 1996 and August 2001. Delirium was defined as a transient mental syndrome of acute onset characterized by global impairment of cognitive functions, a reduced level of consciousness, attentional abnormalities, increased or decreased psychomotor activity, and a disordered sleep-wake cycle. The overall prevalence of postoperative delirium was 8.4%. Of 49 selected patient-related risk factors and treatment variables, 35 were highly associated with postoperative delirium by univariate analysis. Stepwise logistic regression revealed the following variables as independent predictors of delirium: history of cerebrovascular disease, peripheral vascular disease, atrial fibrillation, diabetes mellitus, left ventricular ejection fraction of 30% or less, preoperative cardiogenic shock, urgent operation, intraoperative hemofiltration, operation time of 3 hours or more, and a high perioperative transfusion requirement. Two variables were identified as having a significant protective effect against postoperative delirium: beating-heart surgery and younger patient age. Postoperative delirium is a common complication in cardiac operations. The increased use of beating-heart surgery without cardiopulmonary bypass may lead to a lower prevalence of this complication and thus improve patient outcomes.

Preoperative and operative predictors of delirium after cardiac surgery in elderly patients.

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Delirium is a common complication in elderly patients after cardiac surgery and is associated with adverse outcomes including prolonged hospital stay and increased mortality. Therefore, prevention or early detection of delirium is indicated. Our objective was to identify preoperative and operative characteristics that could predict delirium after cardiac surgery in elderly patients. We conducted a prospective cohort study in which we analysed 201 patients of 70 years and older who underwent cardiac surgery, for developing a delirium. Patients were assessed daily using the Confusion Assessment Method-Intensive Care Unit. Sixty-three patients (31%) developed a delirium after cardiac surgery. The Mini-Mental State Examination (MMSE) score prior to surgery was lower in the delirious patients when

compared with the non-delirious patients (27 vs. 28, $P = 0.026$), creatinine level was higher (98 vs. 88 $\mu\text{mol/l}$, $P = 0.003$) and extracorporeal circulation (ECC) time was longer (145 vs. 113 min, $P < 0.001$). Mortality during the first 30 days after surgery in patients with delirium was significantly higher than that in the non-delirious patients (14 vs. 0%, $P < 0.001$). Low MMSE score and high creatinine level prior to surgery as well as increased ECC time are important independent predictors of delirium. In addition, delirium is an important predictor of 30-day mortality. Patients with a substantial risk for delirium should be candidates for interventions to reduce postoperative delirium and to potentially improve overall surgical outcomes.

Risk and Consequences of Postoperative Delirium in Cardiac Surgery.

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Delirium is a common complication after cardiac surgery that leads to increased costs and worse outcomes. This retrospective study evaluated the potential risk factors and postoperative impact of delirium on cardiac surgery patients. One thousand two hundred six patients who underwent open-heart surgery within a single year were included. Uni- and multivariate analyses of a variety of pre, intra-, and postoperative parameters were performed according to differences between the delirium (D) and nondelirium (ND) groups. The incidence of delirium was 11.6% ($n = 140$). The onset of delirium occurred at 3.35 ± 4.05 postoperative days with a duration of 5.97 ± 5.36 days. There were two important risk factors for postoperative delirium: higher age (D vs. ND, 73.1 ± 9.04 years vs. 69.0 ± 11.1 years, $p < 0.001$) and longer aortic cross-clamp time (D vs. ND, 69.8 ± 49.9 minutes vs. 61.6 ± 53.8 minutes, $p < 0.05$). We found that delirious patients developed significantly more frequent postoperative complications, such as myocardial infarction (MI) (D vs. ND, 1.43% [$n = 3$] vs. 0.28% [$n = 2$], $p = 0.05$), cerebrovascular accident (D vs. ND, 10.7% [$n = 15$] vs. 3.75% [$n = 40$], $p < 0.001$), respiratory complications (D vs. ND, 16.4% [$n = 23$] vs. 5.72% [$n = 61$], $p < 0.001$), and infections (D vs. ND, 36.4% [$n = 51$] vs. 16.0% [$n = 170$], $p < 0.001$). The hospital stay was longer in cases of postoperative delirium (D vs. ND, 23.2 ± 13.6 days vs. 17.4 ± 12.8 days, $p < 0.001$), and fewer patients were discharged home (D vs. ND, 56.0% [$n = 65$] vs. 66.8% [$n = 571$], $p < 0.001$). Because the propensity for delirium-related complications is high after cardiac surgery, a practical, preventative strategy should be developed for patients with perioperative risk factors, including higher age and a longer cross-clamp time.
